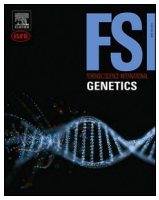
Forensic Science International: Genetics 76 (2025) 103208



Contents lists available at ScienceDirect

Forensic Science International: Genetics

journal homepage: www.elsevier.com/locate/fsigen





Searching national DNA databases with complex DNA profiles: An empirical study using probabilistic genotyping

Séverine Nozownik [a,*], Tacha Hicks [a,b], Patrick Basset [a], Vincent Castella [a]

[a] Unit of Forensic Genetics, University Center of Legal Medicine, Lausanne - Geneva, Lausanne University Hospital and University of Lausanne, Chemin de la Vulliette 4, Lausanne CH - 1000, Switzerland
[b] Fondation pour la formation continue Universitaire Lausannoise (UNIL-EPFL), School of Criminal Justice, Batochime, Dorigny 1015, Switzerland

ARTICLE INFO

Keywords:
DNA Database search
DNA mixtures
Probabilistic genotyping
DBLR
CODIS

ABSTRACT

In most National DNA databases (NDNADB), only single source DNA profiles, and sometimes two-person DNA mixtures, can be searched provided a minimum number of loci (or alleles) is available. DNA profiles that do not meet these criteria (about 14 % of the traces analyzed in Western Switzerland) can be compared locally with candidates upon request from police services, used for one-off search, or remain unused. With the advent of probabilistic genotyping (PG), such complex DNA profiles can be compared to those stored in NDNADB based on likelihood ratios (LRs). In this pilot study, traces of known contributors and casework DNA profiles were used to evaluate the performance of the DBLR™ "Search database" tool in conjunction with the Swiss NDNADB. First, 40 DNA mixtures (2–5 contributors) from 15 volunteers were prepared in the wet laboratory. They were deconvoluted with STRmix™ and compared to a database containing the DNA profiles of these 15 volunteers, along with 174,493 person DNA profiles from the Swiss NDNADB (ground-truth experiments). Using LR thresholds of 10^3 and 10^6, sensitivity and specificity were respectively 90.0 %/57.1 % and 99.9 %/100.0 %. For the lower LR threshold, this resulted in 52 adventitious associations out of more than 24 million pairwise comparisons. Second, 160 DNA mixture profiles from casework (2–4 contributors) that had previously been locally compared were searched with DBLR™ using the same conditions as for phase 1. With the 10^3 LR threshold, 380 associations were retrieved: 194 of these corresponded to expected associations, as they were previously made through the local comparisons with known persons, and 186 were new. With the 10^6 LR threshold, 199 associations were recovered of which 180 were expected and 19 new. This demonstrates that even with complex DNA profiles (up to 4 contributors) all expected associations were retrieved with a limited number of candidates per trace. Database searches of complex DNA mixtures allow for the generation of leads early in an investigation for DNA profiles that might otherwise remain underutilized. Next steps for the possible integration of DBLR™ or similar software within an operational context will require discussions on legal, financial, and technical aspects among stakeholders.

1. Introduction

Most countries have national DNA databases (NDNADB), and many use the program CODIS™ (Combined DNA Index System) to provide investigative leads. The criteria for uploading and comparing DNA profiles in those databases differ between countries and their respective legislations. Generally, all countries allow for the search and storage of single source DNA profiles, provided a minimum number of loci (or of alleles) is available. This rule typically also applies to the major contributor of a DNA mixture. Six is the minimum number of loci that DNA traces must contain for the exchange of DNA data between member countries of the Prüm consortium [1]. In a few countries (e.g., Belgium, Switzerland), two-person DNA mixtures can also be stored and searched, provided they have a minimum number of loci (8 in Switzerland and Belgium).

Many DNA profiles derived from trace DNA collected at the scene, particularly those collected from objects believed to have been touched by the perpetrator(s), do not meet databasing criteria. Most of these profiles are DNA mixtures of more than two persons, partial DNA profiles and minor fractions of DNA mixtures with a CODIS™ compatible major fraction. For simplicity, we refer to these profiles as "complex DNA profiles" in this article. This implies that a significant proportion of

* Corresponding author.
E-mail address: severine.nozownik@chuv.ch (S. Nozownik).

https://doi.org/10.1016/j.fsigen.2024.103208
Received 15 October 2024; Received in revised form 9 December 2024; Accepted 9 December 2024
Available online 10 December 2024
1872-4973/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

DNA profiles are not considered when using the NDNADB. In Western Switzerland, about 15'000 DNA traces were processed in 2023. Among them, only 45 % met the NDNADB introduction criteria: 40 % were single-source DNA profiles (or DNA from a major contributor) and 5 % were two-person DNA mixture profiles. A further 10 % of the DNA traces yielded DNA profiles from more than two persons, which were discriminative and valuable, although they could not be entered into the NDNADB. Additionally, about 4 % of the DNA traces had exploitable minor DNA fractions. Combined, this represents more than 2'000 DNA traces that do not meet the Swiss NDNADB criteria but could provide investigative leads. Finally, about 45 % of the traces analyzed had DNA profiles that were deemed "not interpretable", generally when the allelic content was not reproducible between replicate analyses, or "no profile" when fewer than five alleles were present on the electropherograms (EPG) (at 16 loci).

In Switzerland, complex DNA profiles can currently be exploited using two processes. The first involves comparing, but not storing, the profiles in the NDNADB. These are called one-off or speculative searches and have shown to be of value to the police. In 2023, we conducted 115 one-off searches. The other process involves the "manual" comparison of DNA profiles without using the NDNADB. In Western Switzerland, these are known as local comparisons: they are performed routinely but only on request. They require the police to ask for specific comparisons either between DNA profiles of traces or between a trace and one or several persons of interest (POI) suggested by the investigation. In 2023, 1146 local comparisons were conducted in our laboratory. Despite these processes, many complex interpretable DNA profiles remain unused.

Indeed CODIS™, the software that is used in most countries for database search, is not well-suited for comparing complex DNA profiles. It does not utilize peak heights, or account for stochastic variations in DNA profiles (e.g., peak imbalances, drop-outs or drop-ins). Consequently, partial DNA profiles are only loaded when the allelic content of a locus is considered 100 % reliable. When comparing DNA profiles, CODIS™ considers the presence or absence of matching alleles at the selected loci, resulting in a loss of discriminating power. Moreover, CODIS™ does not generally account for the rarity of a genotype, nor does it compute likelihood ratios (LRs) to assign the value of the DNA comparison. Probabilistic Genotyping (PG) was developed in response to the increasing complexity of DNA profiles [2,3]. PG software considers the height of the peaks present at all the loci analyzed, incorporates stochastic effects (for a review see [3,4]) and allows to assign LRs. LRs have been shown to perform better as a metric for comparing DNA profiles, also in the context of NDNADB [5,6], compared to the match stringency used by CODIS™. This issue is even more pronounced for DNA mixtures. Several software implementing continuous models, such as DBLR™ [7] and ProbRank™ [8], allow for the rapid calculation of LRs to compare DNA profiles of traces to those of persons present in a database. Qualitative models have also been used for database purposes, such as SmartRank™ [9] and DNAmatch2™ [10]. These comparisons are based on the LR. As a LR is computed for all candidates, they can be ranked accordingly [9,10].

Studies evaluating database search tools based on PG (or otherwise) are limited. Bright et al. [11] performed a study using STRmix™ and the New-Zealand database to assess the usefulness of searches of unresolved mixtures. Benschop et al. [9] first evaluated the performance of SmartRank™ DNA database search by comparing 343 computer-simulated DNA mixture profiles with several European NDNADB, and second, compared DNA database search performance between SmartRank™ and CODIS™. Hoogenboom et al. [8] demonstrated the potential of ProbRank™ in the Netherlands. Bright et al. [11] compared casework DNA mixture profiles with a real NDNADB, however the number of profiles was limited to two.

The aim of this research is to evaluate the DBLR™ "Search database" tool on both 40 ground-truth experimental traces and 160 casework DNA mixtures to assess its value in an operational context. To our knowledge, this is the first study relying on the comparison of a large number (160) of complex casework DNA profiles to a real NDNADB.

2. Material & methods

2.1. Overall description

Access to the anonymized DNA profiles of the individuals in the NDNADB for this research project was granted by fedpol (Swiss Federal Office of Police), the custodian of the database. The DBLR™ team provided a script to input the data of the DNA profiles of the 174,493 individuals present in the NDNADB at the time of the study (13/03/2023) into DBLR™. The persons' DNA profiles present in the NDNADB were analyzed with different amplification kits: about 78 % have 16 loci, 18 % have 10 loci and 4 % have 11 loci. In theory, the number of adventitious associations should decrease when increasing the number of loci, but this was not studied here.

The project was divided into two parts. First, the DNA from 15 volunteers (11 women and 4 men), who signed an informed consent form, was used to construct ground-truth DNA mixture profiles with known characteristics (ground-truth experiments; see below). These were analyzed using the same chemistry applied in casework. The DNA mixture profiles were deconvoluted with STRmix™ [12] using the parameters established during implementation, and then searched in DBLR™ against the DNA profiles of the 174,493 persons and the 15 volunteers. This part was designed to assess the specificity and the sensitivity of the approach on the mixtures tested.

In the second part of this study, 160 DNA mixture profiles from previous casework were searched against the same individuals to replicate casework conditions. Following a request from fedpol, the resulting data were not shared further (neither with the volunteers nor with the police), regardless of the results obtained.

2.2. Ground-truth study: known DNA mixtures

Using DNA extracts from the buccal swabs of the 15 volunteers, 40 mixtures with different number of contributors (2−5) were prepared in the laboratory. The DNA from the buccal swabs (4N6 FLOQSwabs Genetics, Copan) was extracted using the QiAmp® DNA Minikit (Qiagen, Germany) and quantified using the Quantifiler™ Trio DNA Quantification kit (Thermo Fisher Scientific, Reinach, Switzerland). Dilutions were then prepared for each of these 15 DNA extracts to obtain final DNA concentrations of 100 pg/µl, 50 pg/µl and 20 pg/µl. The 40 DNA mixtures (Table 1) of two persons (2 P), 3 P, 4 P and 5 P were then prepared to obtain a final total DNA concentration of 200 pg/µl. The DNA mixtures were designed, with half of them having a balanced ratio and the other half an unbalanced ratio, to mimic casework situations.

Two amplifications of 30 PCR cycles were performed for each DNA mixture using the AmpFLSTR® NGM SElect™ PCR Amplification kit (Thermo Fisher Scientific, Reinach, Switzerland) with 200 pg of template DNA on a Veriti™ 96-well Fast Thermal Cycler (Thermo Fisher Scientific, USA). The resulting amplicons were then run on a 3500 Genetic Analyzer (Thermo Fisher Scientific, USA). The DNA mixture profiles were analyzed with the GeneMapper™ software (Thermo Fisher

Table 1
Ratio of the balanced and unbalanced DNA mixtures of 2 P, 3 P, 4 P and 5 P prepared in the laboratory.

DNA mixtures	Number of mixtures	Ratio	DNA per contributor [pg]
2 P	5	1: 1	100: 100
	5	9: 1	180: 20
3 P	5	1: 1: 1	67: 67: 67
	5	6: 3: 1	120: 60: 20
4 P	5	1: 1: 1: 1	50: 50: 50: 50
	5	4: 3: 2: 1	80: 60: 40: 20
5 P	5	1: 1: 1: 1: 1	40: 40: 40: 40: 40
	5	4: 2: 2: 1: 1	80: 40: 40: 20: 20

Scientific, USA). Each profile was subsequently deconvoluted using STRmix™ (v2.7 for 2/3/4 P, v2.9 for 5 P). Two different versions (v2.7 and v2.9) were used to deconvolute the DNA mixture profiles, as the experiments coincided with the in-house verification of the newest version, which enabled deconvolution of 5-person mixtures with our specifications (Intel® Core™ i9–10920X CPU 3.50 GHz, RAM: 128Go, 12 Core, 64-bit, Windows 10). Both PCR replicates were used for the deconvolution within STRmix™ and in the subsequent DBLR search. The deconvolution time depends on the DNA profile and the number of contributors to the mixture: maximum 2 minutes for 2- and 3- person mixtures, up to 50 minutes for 4-person mixtures and maximum 2 hours for 5-person mixtures. Once the deconvolutions were performed, the DBLR™ database search only took less than 1 minute, with a database our size, no matter the number of contributors to the mixture. This run time was considered adequate for our daily practice. Note that it is possible to have multiple licenses or computers with higher specifications to reduce processing time.

2.3. Casework DNA mixtures

A total of 160 DNA mixtures from casework were selected from those analyzed and deconvoluted in 2022, using STRmix™ version 2.7. These DNA profiles were previously deconvoluted and manually compared to given candidates (whose DNA profiles were present in the NDNADB) and the corresponding LRs were computed. At that time, in our laboratory, the maximum assigned number of contributors for casework DNA mixtures was 4, as the validation did not include 5 person DNA mixtures. Since we already had LRs for these mixtures when compared to their given candidate, we expected to observe the same leads with DBLR™. Among the 160 traces, 50 were assigned as two-person DNA mixtures, 50 were as 3-person mixtures and 60 as 4-person mixtures. The LRs of the selected potential contributors ranged from 10^5 to 10^{30} for the two-person DNA mixtures, from 10^4 to 10^{25} for the 3-person and from 10^0 to 10^{23} for the 4-person mixtures. With a LR threshold of 10^3, we expected to retrieve leads for 157 traces (3 mixtures assigned as 4-person were below threshold). Similarly, with a LR threshold of a million, we expected to retrieve the given candidates for 150 traces (50 were assigned as 2 P mixtures, 49 as 3 P and 51 as 4 P).

2.4. DBLR™ database searches

DBLR™ (v1.2.24) was used for both experiments. Database (DB) searches were performed using the DBLR™ "Search Database" tool. First, the 40 DNA mixtures of known contributors (ground truth study) were compared to the 174,493 person DNA profiles present in the NDNADB, as well as the 15 DNA profiles of the volunteers. Second, 160 DNA mixtures of respectively 2 P, 3 P and 4 P from casework previously deconvoluted were compared to the same persons.

The LRs were computed for each person of the database given the following propositions:

- The DNA originated from the candidate and *N*-1 unknown, unrelated individuals.
- The DNA originated from *N* unknown individuals unrelated to the candidate.

N represents the experimentally designed number of contributors. When the DNA mixtures were conditioned in casework, meaning that a person's DNA contribution was assumed during deconvolution, the conditioning was retained, so in some mixtures the propositions considered were:

- The DNA originated from the candidate, the complainant and *N*-1 unknown, unrelated individuals.
- The DNA originated from the complainant and *N* unknown individuals unrelated to the candidate.

The Swiss population data [13] and a fixation index (F_{ST}) of 1 % [14] were used. All contributors' positions were considered: for each search, the candidate was considered as a potential contributor for all parts of the mixture (from contributor 1 to the largest number of contributors of the mixture). Thus, we considered sub-source, not sub-sub source propositions [15].

LR thresholds ranging from 1 to 10^9 were used for the ground-truth DNA mixtures. For the DNA mixtures from casework, LR thresholds of 10^3 and 10^6 were applied. The 10^6 LR threshold is comparable to the expected LR obtained for DNA comparisons involving single source DNA profiles that share a minimum of 6 loci with a potential candidate. This threshold is used in several countries such as Switzerland, France and Belgium and has been employed in various publications [11,16,17]. This threshold could be used in routine casework. The 10^3 LR threshold was used as an example of a lower threshold that could be applied in special cases.

3. Results

3.1. Experiment 1: DNA database searches with DNA mixtures of known contributors

The aim of the first part of our study was to determine the sensitivity and the specificity of the DBLR™ "Search database" tool using our ground-truth DNA mixtures with 2–5 known contributors. A LR threshold of 1 was applied to provide a global view of the LRs obtained for the 24 million pairwise comparisons conducted (i.e., 40 mixtures of 2, 3, 4 and 5 contributors compared with 174,508 persons).

3.1.1. Comparison with known contributors

All comparisons between the DNA mixtures and their known contributors were characterized by LRs$>$ 1. As expected, the LRs obtained for known contributors decreased with the increasing number of contributors (see Fig. 1). For two-person DNA mixtures, all LRs were above 10^9. For 3-person DNA mixtures the lowest LR reported was 10^3, with 90 % of the LR greater than or equal to 10^6. For 4- and 5-person mixtures, the LRs were smaller (as expected due to the reduced information content) with some being below to 10^3. However, for 4-person mixtures (respectively 5-person), 53 % (24 %) of the known contributors had a LR greater than or equal to 10^6.

Using a 10^3 LR threshold, 126 out of 140 (90 %) contributors were retrieved from the database. Only 14 out of 140 (10 %) contributors were not indicated as potential contributors, all of which concerned 4-person (4) or 5-person (10) mixtures. With a LR threshold of 10^6, 80 out of 140 (57 %) contributors were retrieved. LRs bellow 10^6 were mostly computed for the comparisons with 5-person DNA mixtures (38 LRs) and 4-person mixtures (19 LRs); for the 3-person mixtures, only 3 comparisons had LRs smaller than 10^6.

In addition to the number of contributors, the mixture proportion and the DNA amount per contributor impact the results. Fig. 2 illustrates that, given peak height information differs, very large LRs can be obtained even for 5-person mixtures. When comparing the DNA amount (computed by STRmix™ in RFU) with the LR, it is observed that increasing the DNA amount leads to larger LRs for known contributors in both balanced and unbalanced mixtures (see Fig. 2). Conversely, when the DNA amount decreases, the LR of known contributors tends toward 1 (Log10(LR)= 0), especially for 5-person DNA mixtures, regardless of whether the mixture is balanced or unbalanced (see Fig. 2). This trend is expected, as the information given by the electropherogram (EPG) for that contributor is significantly reduced.

3.1.2. Comparison with non-contributors

Depending on the threshold adopted for the database search, adventitious associations were observed when performing a DBLR™ database search. However, most non contributors were assigned a LR smaller than one (99.8 %). The results showed that the number of

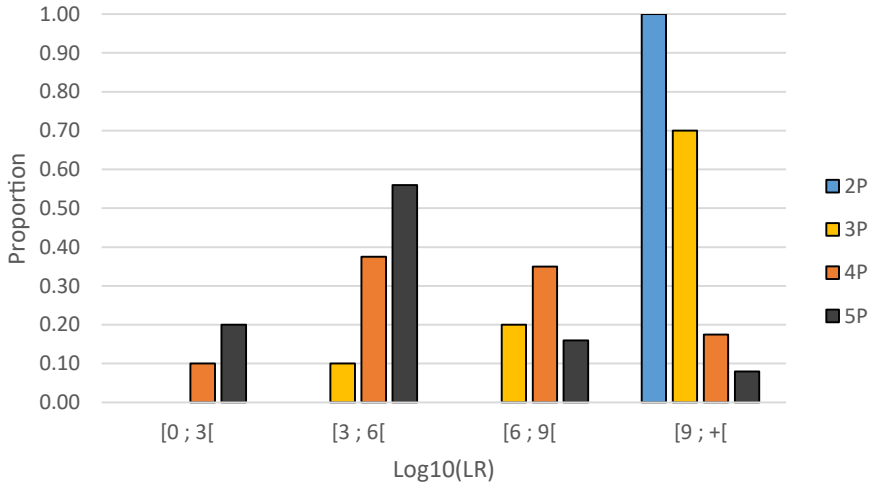


Fig. 1. Log10 LRs obtained for known contributors in ground-truth DNA mixtures of 2, 3, 4 and 5 persons (N = 140 for the 40 mixtures).

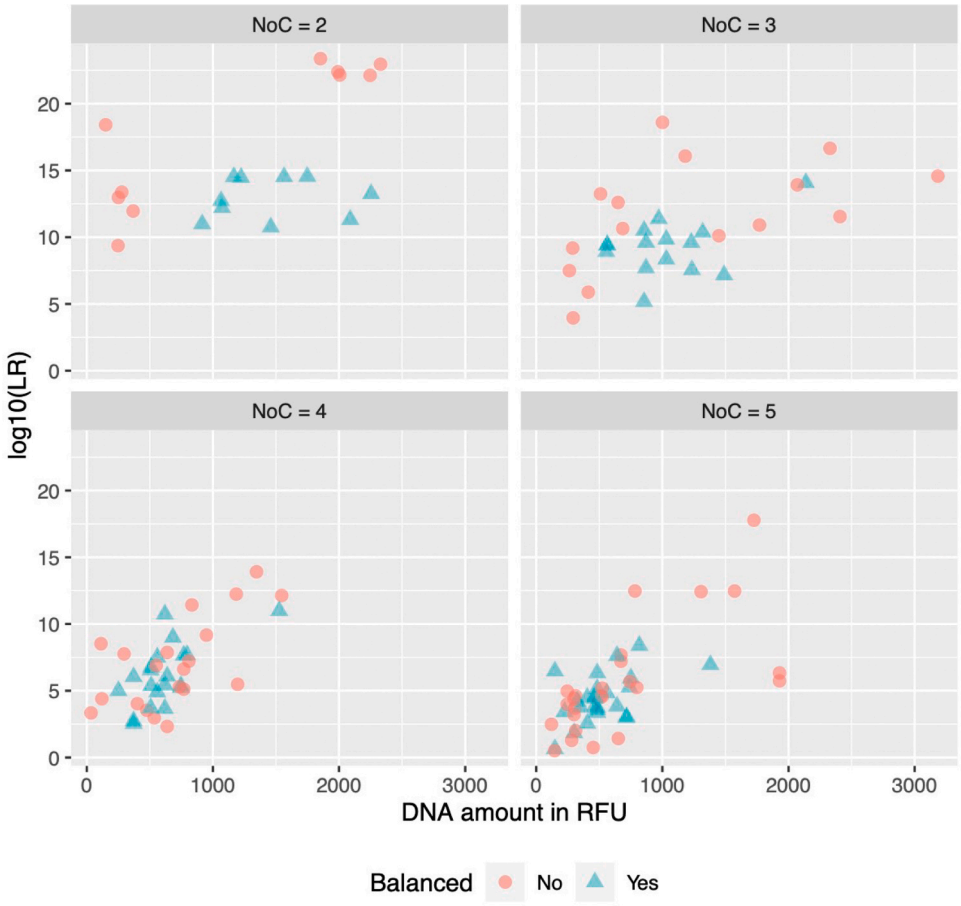


Fig. 2. Log10(LRs) obtained when comparing the DNA mixtures constructed with 2, 3, 4 and 5 persons (N = 140) with the known contributors. The DNA amount (in RFU) and whether the mixture is balanced or unbalanced are indicated.

adventitious associations increased with the number of contributors to the mixture (see Table 2), and as expected, when the LR threshold was low. For two-person DNA mixtures, one adventitious association on 137,350 comparisons was observed from the database search with a LR threshold of 1. In contrast, for 5-person mixtures this number increased to 11,340. This increase is expected due to the smaller information content. In our experiments, no adventitious associations were observed for 2- and 3-person mixtures, as well as for 4- and 5-person DNA mixtures, when the LR thresholds were set at $\geq 10^4$ and $\geq 10^5$ respectively. With a LR threshold of 10^3, the software returned 52 adventitious associations, of which only 3 were for 2- and 3-person DNA mixtures.

The number of candidates proposed per mixture depended on the number of contributors. Using the 10^3 threshold, the maximum number of candidates obtained for a two-person DNA mixture was 3, with a median of 2. For a 5-person DNA mixture, the maximum number was 9 with a median of 7 (see Table 3). With the 10^6 threshold, the maximum

Table 2
Number of adventitious associations obtained in the ground-truth mixtures experiment using different LR thresholds and a database of N = 174,508 persons' DNA profiles, including the contributors of these mixtures. True contributors were not considered as adventitious.

Log10(LR) threshold	2 P	3 P	4 P	5 P	Total number of adventitious associations
0	1	42	2352	11340	13735
1	1	24	712	2224	2961
2	1	8	152	279	440
3	1	2	20	29	52
4	0	0	2	2	4
5	0	0	0	0	0
6	0	0	0	0	0

Table 3
Total number of candidates (including true and false leads) obtained per mixture, depending on the number of contributors (P) to the DNA mixtures in the ground-truth mixtures experiment. Results are shown using a LR threshold of 10^3 (10^6) and a database of N = 174,508 persons' DNA profiles, including the contributors to these mixtures.

Number of candidates	2 P	3 P	4 P	5 P
Minimum	2 (2)	3 (2)	3 (1)	5 (0)
Maximum	3 (2)	4 (3)	9 (3)	9 (3)
Median	2 (2)	3 (3)	5 (2)	7 (1)

number of candidates for two-person DNA mixtures was 2, with an identical median. For 5-person DNA mixtures, the maximum number of candidates returned was 3, with a median of 1.

For 2-person and 3-person mixtures, known contributors always yielded a larger LR than the adventitious matches. This was not the case for 4-person and 5-person mixtures where known contributors and adventitious matches LRs overlapped for some DNA mixtures.

We used ground-truth experiments (with both true contributors and non-contributors) to explore sensitivity (the software's ability to reliably resolve the genotype(s) of a DNA profile's contributor(s) for a range of profiles) and specificity (the software's ability to reliably exclude non-contributors within a DNA profile for a range of starting DNA templates). More specifically, for the sensitivity, we calculated the probability of observing a LR larger than a given threshold when the DNA originated from the candidate and N-1 unknown, unrelated individuals. For the specificity, we used one minus the probability of observing a LR larger than a given threshold when the DNA originated from N unknown individuals unrelated to the candidate. We used LR thresholds ranging from 1 to 10^9 (see Table 4). As expected, sensitivity decreased with increasing LR thresholds, while the specificity increased. In our dataset, the highest specificity for the mixtures tested was achieved with LRs $\geq 10^5$ (100 %).

Table 4
Sensitivity and specificity were calculated based on the comparisons between 40 ground-truth DNA mixtures of 2, 3, 4 and 5 known contributors and 174,508 persons' DNA profiles (including the contributors to the mixtures) using different LR thresholds in the context of a database search.

Log10LR	Sensitivity	Specificity
0	100.00 %	99.80 %
1	97.86 %	99.96 %
2	95.71 %	99.99 %
3	90.00 %	99.99 %
4	78.57 %	99.99 %
5	69.29 %	100.00 %
6	57.14 %	100.00 %
7	50.00 %	100.00 %
8	40.71 %	100.00 %
9	37.14 %	100.00 %

3.2. Experiment 2: DNA database searches with casework DNA mixtures

The aim in this second part was to assess the potential of the DBLR™ "Search database" tool under casework conditions. Previous local comparisons had been conducted with DNA mixtures that were not suitable for searching with CODIS™, but whose potential candidates were in the Swiss national DNA database. This experiment investigated whether these associations were retrieved by DBLR™ as expected, how many candidates would be proposed, and the rank of our local candidates. Specifically, 2-, 3-, 4-person DNA mixtures from casework were compared against the NDNADB and the 15 reference profiles from volunteers. A defined number of candidates were expected from the selected traces (see Table 5 & Table 7).

3.2.1. DBLR results when setting a LR threshold at one million

To better approximate realistic criteria and to test the DBLR™ "Search database" tool in conditions close to our current ones, the results were analyzed using a LR threshold of 10^6.

With this threshold, 199 candidates were suggested for 150 out of 160 traces considered (see Table 5). All 180 previously observed associations were retrieved, provided that the candidates had not been removed from the NDNADB between the local comparison and the export of the individuals' DNA profiles into DBLR™. Additionally, 19 new leads were observed. Note that 5 expected associations were not retrieved. These cases correspond to situations where the individual's DNA profile was removed from the NDNADB between the time of the local comparison and the current project, in accordance with the Swiss regulations [18]. These 5 persons' profiles were thus not present in the 174,508 persons' DNA profiles. Since these DNA profiles had been removed from the database, these cases were not considered as "fails".

As observed in the first part based on ground truth experiments with known contributors, the LRs varied depending on the number of contributors to the DNA mixtures (see Fig. 3). Specifically, the LRs tended to decrease as the number of contributors increased. The LR assigned with DBLR™ were the same as the ones previously assigned with STRmix™ at the time of the local comparison.

With a LR threshold of 10^6, 19 new candidates were suggested (see Fig. 4). There were 1, 6 and 12 associations for 2, 3 and 4-person DNA mixtures, respectively. Most of these associations (18) involved DNA mixture profiles of 3 persons or more, and 15 of these (78.9 %) had a LR of the order of a billion. The LR was of the order of 10^9 for the new single candidate for the two-person DNA mixtures, for the 4 new leads regarding of 3-person DNA mixtures, and for the 10 new leads from 4-person DNA mixtures. It should be noted that the 2-person mixture had not been locally compared as it was part of a series of traces sharing similar allele contents.

For two-person DNA mixtures either 1 or 2 candidates were returned by the DBLR™ search. For both 3-person DNA mixtures and 4-person DNA mixtures, in addition to the associations with 1 or 2 candidates, 3 candidates were observed for one DNA mixture profile (see Table 6).

Table 5
Leads obtained following the DBLR™ database search with a 10^6 LR threshold using DNA traces from casework (N = 150/160) and a database of 174,508 persons' DNA profiles.

	Local comparisons	DBLR™ « Search database »	
	Associations observed following local comparison	All leads suggested using DBLR™*	"New" leads using DBLR™
2 P	71	72	1
3 P	55	61	6
4 P	54	66	12
Total	180	199	19

* *All expected associations observed in local comparisons with a 10^6 threshold and new leads (e.g., 71 +1 =72).*

Table 6
Total number of candidates obtained per mixture with a lead (N = 150), depending on the number of contributors to the DNA mixtures in the casework mixtures experiment using a LR threshold of 10^6 and a database of N = 174,508 persons' DNA profiles.

Number of mixtures	2 P	3 P	4 P
With one candidate	28	38	37
With two candidates	22	10	13
With three candidates	0	1	1
Total	50	49	51

Table 7
Leads obtained from the DBLR™ database search with a 10^3 LR threshold using DNA traces from casework (N = 157/160) and a database of 174,508 persons' DNA profiles.

	Local comparisons	DBLR™ « Search database »	
	Associations observed following local comparison	All leads suggested using DBLR™*	"New" leads using DBLR™
2 P	72	79	7
3 P	56	118	62
4 P	66	183	117
Total	194	380	186

* *All expected associations observed in local comparisons with a 10^3 threshold and new leads (e.g., 72 +7 =79).*

3.2.2. DBLR results when setting a LR threshold at 1000

A LR threshold of 10^3 was used because it provided a good balance between sensitivity and specificity in part 1 of this research, and because the number of candidates per profile with a threshold of one million was manageable.

With 10^3 LR threshold, 380 associations were highlighted for 157 out of 160 DNA casework mixtures (see Table 7). Of those, 194 (51 %) corresponded to the expected associations that were previously observed in local comparisons. Additionally, 186 new associations were obtained. Similar to the 10^6 LR threshold, 5 expected associations were not retrieved because the DNA profile of the person of interest had been removed from the NDNADB between the local comparison and the present project.

Omitting the candidates removed from the NDNADB, all expected associations observed in previous local comparisons were retrieved with the DBLR™ search using the 10^3 LR threshold (Fig. 5). Most of the new associations were observed for 4-person mixtures (see Table 7).

With this 10^3 cut-off, 186 new associations were revealed (compared to the 19 with a LR threshold of a million). Thus, most of these new possible leads (80 %) had LRs < 10^6 (Fig. 6). This contrasts with what was observed with local comparison (Fig. 5 compared to Fig. 6). Indeed, local comparisons are only conducted with candidates designated by the investigation which is not the case for "blind" NDNADB searches.

For most two-person DNA mixtures (49 out of 50), 1 or 2 candidates were returned by the DBLR™ search and 1 candidate was returned with 7 associations. For 3-person DNA mixtures, 32 out of 50 were associated with 1 or 2 candidates and the 18 remaining with 3–7 associations. For 4-person DNA mixtures, 26 out of 57 were associated with 1 or 2 candidates and for the 31 remaining mixtures, 3–12 candidates were proposed (see Table 8).

4. Discussion

DNA profiles considered for routine searches within national DNA databases (NDNADB) are generally single-source DNA profiles or, in some countries, such as Belgium and Switzerland, DNA mixtures from two contributors. Traditional software such as CODIS™ (Combined DNA Index System) relies solely on the presence or absence of reported alleles to compare DNA profiles. While this approach is efficient for high-quality single-source DNA profiles, using peak heights in DNA mixtures has been shown to improve both sensitivity and specificity [19]. This enhancement may resonate with DNA scientists who, when manually comparing a trace DNA mixture to individual profiles, exclude potential candidates when the peak heights in the electropherogram do not align with expectations. Probabilistic genotyping (PG), however, incorporates peak height information along with factors such as drop-out and drop-in probabilities, DNA degradation, locus amplification efficiency, replicate variability and DNA template quantity [4]. As a result, the manual exclusion process described above can be automated, even for complex DNA profiles involving two or more contributors. This study aimed to evaluate the potential of DBLR, in combination with PG software, to reveal associations between DNA mixtures from up to five contributors and a database containing the DNA profiles of 174,493 individuals' profiles from the Swiss NDNADB as well as 15 volunteer profiles, for a total of 174,508 profiles.

In the first part of the study, 40 ground-truth DNA mixtures from two to five volunteers, both balanced and unbalanced, were compared to the 174,508 person DNA profiles. This analysis highlighted that the characteristics of the mixture, such as the number of contributors, mixture proportion and DNA amount per contributor, affect the magnitude of the LR, as previously reported [20]. For instance, DNA mixtures with a greater number of contributors tend to show lower LRs (see Fig. 1), and

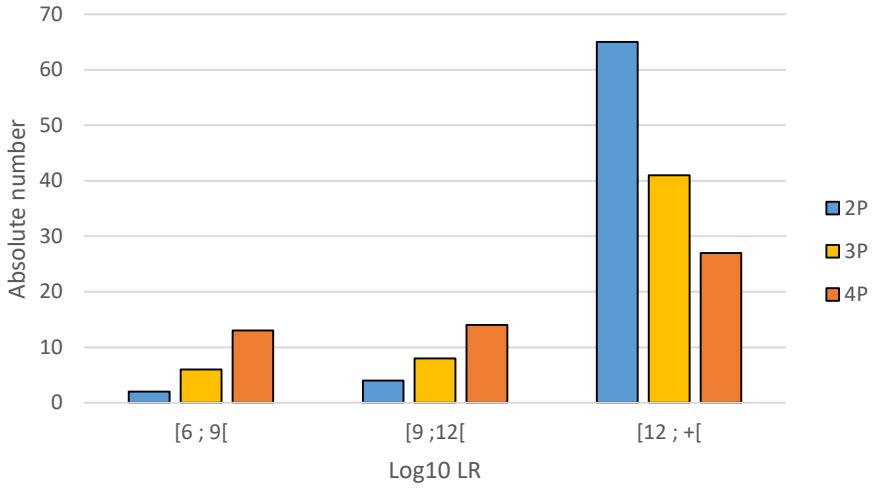


Fig. 3. Log10(LR) distribution of the expected leads (N = 180 following local comparisons) obtained with the DBLR™ database search for DNA mixture profiles from casework of 2, 3, 4 persons with a LR threshold of 10^6.

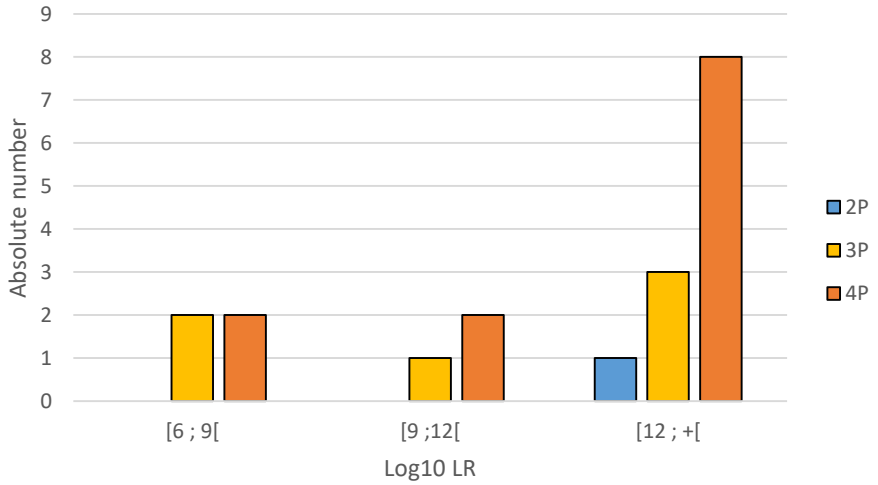


Fig. 4. Log10(LRs) obtained for the new leads (N = 19) using the DBLR™ database search for DNA mixture profiles from casework of 2, 3, 4 persons with a LR threshold of 10^6.

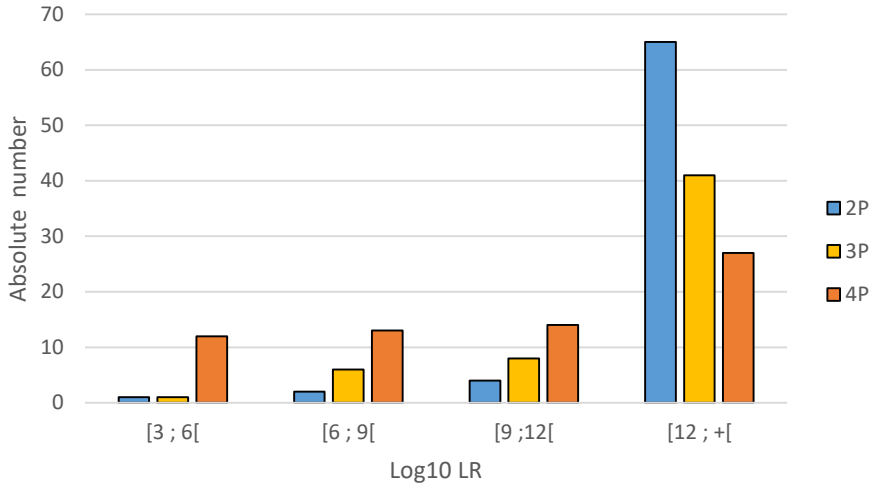


Fig. 5. Log10(LR) distribution of the expected leads (N = 194 following local comparisons) obtained with the DBLR™ database search for DNA mixture profiles from casework of 2, 3, 4 persons with a LR threshold of 10^3.

DNA profiles with lower DNA amount also generally had lower LRs (see Fig. 2). Despite this, large LRs were observed for several known contributors, even for 5-person mixtures, particularly for major contributors in unbalanced DNA mixtures. This suggests that DBLR™ can efficiently retrieve true contributors even with complex DNA profiles, although the mixture profile characteristics and the LR threshold used for the DBLR™ search will influence the detection of true contributors in known mixtures.

Our ground-truth analysis also demonstrated how sensitivity and specificity varied with the LR threshold used for the search. Our findings align with validation studies of DBLR™ [7] and STRmix™ [12,20]. Using a LR threshold of 1, all expected associations between the DNA mixtures and the volunteers were retrieved. However, this low threshold resulted in 13,735 adventitious associations (see Table 2). In contrast, with a LR threshold of 10^6, no adventitious associations were found, but only 57 % of the expected contributors were retrieved. DNA profiles association with LR greater than 10^6 were recovered for 100 %, 90 %, 53 % and 24 % of DNA mixtures with 2, 3, 4 and 5 contributors, respectively. This indicates that DNA profiles that currently do not meet the criteria for uploading in CODIS™ still have significant potential for generating valuable investigative leads.

The choice of the LR threshold for a search involves a balance between sensitivity and specificity, but other factors, such as the severity of the offence and the available resources, must also be considered. Thus, a universal threshold for all DBLR™ searches cannot be defined. Instead, the threshold should be adapted to the specific circumstances encountered. This could be formalized by decision analysis [21]. For routine comparisons of DNA profiles, a relatively high LR threshold may be appropriate to maximize investigative leads while limiting the resources needed to handle a large volume of potential candidates. In Switzerland, single-source DNA profiles can be uploaded to the NDNADB if they include at least 6 loci, a criterion also used in France or Belgium. An association with such a partial DNA profile would correspond to a minimum LR of 10^6 [5]. This threshold aligns with those used in other database search studies [11,16,17]. For the standard Swiss context, where burglaries and thefts are the most common offenses for which DNA profiling is used [22], this LR threshold seems reasonable. While some associations might be missed with this threshold, the number of proposed candidates remains low (0 or 1 candidates per profile; see Table 3), reducing the time needed to manage these associations. In more serious cases, such as sexual assaults or homicides, lowering the LR threshold might be beneficial to generate leads in absence of other information. This could increase the number of adventitious associations, but our results suggest that this increase

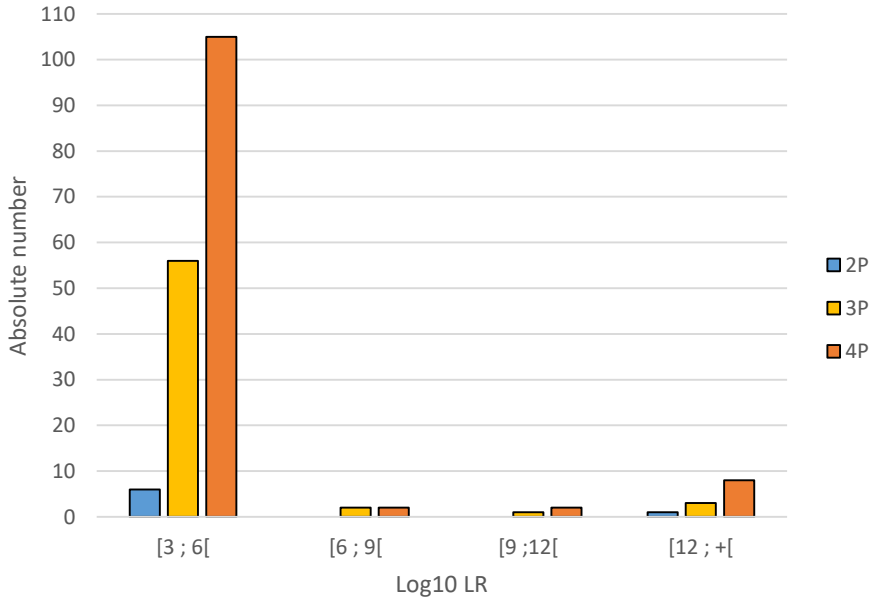


Fig. 6. Log10(LRs) obtained for the new leads (N = 186) using the DBLR™ database search for DNA mixture profiles from casework of 2, 3, 4 persons with a LR threshold of 10^3.

Table 8
Total number of candidates obtained per mixture with a lead (N = 157/160) depending on the number of contributors to the DNA mixtures in the casework experiment using a LR threshold of 10^3 and a database of N = 174,508 persons' DNA profiles.

Number of mixtures	2 P	3 P	4 P
With one candidate	26	19	13
With two candidates	23	13	13
With three candidates	0	7	12
With four candidates	0	6	4
With five candidates	0	3	8
With six candidates	0	1	3
With seven candidates	1	1	2
With eight candidates	0	0	1
With nine candidates	0	0	0
With ten candidates	0	0	0
With eleven candidates	0	0	0
With twelve candidates	0	0	1
Total	50	50	57

would be moderate. For example, the maximum number of candidates per DNA mixture profile was 9 in our ground truth experiment (12 in the casework experiment) using a relatively low LR threshold of 10^3 (see Tables 3 and 8). Such a list of candidates can likely be managed by investigators, especially for serious cases. While lower LR thresholds may produce more false leads, this is not problematic if the database search is used for investigative purposes. Police and prosecutors should keep in mind that DNA evidence is just one piece of the puzzle and must be considered in the context of the case, as recommended by the ISFG commission in guidelines [23,24] and the Expert Working Group on Human Factors in Forensic DNA Interpretation [15]. It is also important to note that the figures reported in this study were based on a database containing 174,508 person DNA profiles. A larger database would likely yield more associations [9]. Therefore, the size of the NDNADB must also be considered when defining an appropriate LR threshold.

In the second part of the study, we evaluated the DBLR™ software in an operational context by comparing 160 casework DNA mixture profiles from two to five contributors with the 174,508 individual profiles in the database. This evaluation was conducted with LR thresholds of 10^3 and 10^6. These DNA profiles had previously been compared with persons of interest suggested by the investigation, a process referred as "local comparisons", at the request of the police. All expected associations from these local comparisons (i.e., associations showing LRs greater than one) were successfully retrieved with DBLR™ using both LR thresholds. This suggests that DBLR™ is an effective alternative to local comparisons, as it allows for direct comparison with the full database rather than a limited number of investigator-suggested candidates. As a result, 186 new associations were retrieved with the 10^3 LR threshold and 19 with the 10^6 LR threshold (see Tables 5 and 7), suggesting that associations not previously highlighted by investigation could be quickly uncovered through direct comparison to the NDNADB.

Using DNA mixtures for NDNADB searches opens up the possibility of generating multiple leads from a single trace. Among the 160 casework DNA mixtures, 47 were associated with two or three persons, with LRs varying depending on the information content of the DNA mixture profiles. Since these mixtures are from casework, we do not know whether these associations are true or false. Traditional searches in CODIS™ may also generate associations with LRs of the order of 10^6 for partial single-source DNA profiles and these leads are considered by police services. There is no reason why DNA mixtures of 2, 3, 4 and 5 persons with similar or larger LR values should not be considered. One advantage of LR-based database search using probabilistic genotyping (PG) software is its ability to compute LRs for all candidates [8–10], allowing for prioritization among them. This contrasts with CODIS™, which provides a list of potential candidates but does not indicate the value of the comparison.

The number of leads observed in the second part of our study might be overestimated, as the DNA mixtures from casework were selected from a pool of DNA profiles deemed suitable for local comparisons. However, our results confirm that DBLR™ (or similar software) represents an efficient tool for comparing currently underused complex DNA mixture profiles to a NDNADB. A previous study reached the same conclusion, albeit on a smaller scale [11]. In Western Switzerland, the systematic use of PG for NDNADB search could increase the number of trace DNA profiles uploaded by approximatively 14 %, representing 2000 DNA profiles per year, most of which are mixtures involving more than two contributors that are not currently searched. In 2023, 33 % of the trace DNA profiles submitted to CODIS™ were associated with individual DNA profiles. Assuming a similar proportion for complex DNA profiles, which is likely conservative because one DNA mixture can potentially generate multiple associations, over 600 new investigative

leads (2000 *33 %) could be provided to the authorities each year. In countries where only single-source DNA profiles are uploaded into the NDNADB, the increase in searchable DNA profiles could exceed 14 %.

While the use of DBLR™ (or similar software) for NDNADB search is promising, it currently has some limitations. Dedicated resources are needed to purchase, install and validate the software. Additionally, whereas unambiguous single-source profile can directly be loaded in the DBLR™ search module, DNA mixture profiles must be specifically processed and deconvoluted before being used in DBLR™. This involves separating the component DNA genotypes of contributors based on peak height information and other factors (e.g., the number of contributors, mixture proportions, or known contributors). These steps can add significant workload for laboratories, especially when many DNA profiles require processing. As a result, deconvolutions are often performed only when necessary for determining the value of a DNA comparison. For example, in our laboratory, DNA mixture profiles are deconvoluted only when required for comparison purposes. Recent developments in automation, such as automatic allele/stutter calling, the preparation of input files and deconvolution batches, is promising and could provide a solution to this issue [25].

5. Conclusions

This study demonstrates that database searches combined with probabilistic genotyping (PG) are effective for conducting NDNADB search with DNA mixtures involving up to five contributors. This contrasts with most laboratories current practice of searching trace DNA profiles from only one and, within some countries, two contributors. This capability allows law enforcement to utilize a greater number of DNA profiles, thereby providing more investigative leads. However, integrating PG into the CODIS™ system will take time. An alternative approach is to run two parallel processes: the existing CODIS™ system for single-source DNA profiles and DBLR™ (or similar software) for DNA mixtures. This process is used for example in Australia and Ireland. Profiles processed with PG could be selected based on a cost-benefit analysis that considers factors such as the number of contributors, the severity of the cases and resource availability. For instance, Australia uses a criterion based on case gravity and expected LRs to select DNA mixture profiles for comparison to the NDNADB with DBLR™ (D. Taylor, personal communication). Before implementing PG in conjunction with NDNADB, legal, financial, and technical considerations must be addressed. In Switzerland, legal issues should not pose a problem, as the comparison of complex DNA profiles is allowed. Nevertheless, discussions are necessary to determine how DBLR™ can be integrated with CODIS™ and to assess their complementarity. While using different software for different types of the DNA profiles may be seen as a temporary solution, future efforts should focus on integrating PG capabilities directly into CODIS™ to maximize the investigation leads generated from currently underutilized DNA profiles.

Funding

The work of SN was supported by the Swiss National Science Foundation (SNSF) through Grant No. 310030E_205558/1 awarded to Vincent Castella.

CRediT authorship contribution statement

Patrick Basset: Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Tacha Hicks:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Séverine Nozownik:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Vincent Castella:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to express our gratitude to the Swiss Federal Police and the DBLR™ team whose support was essential for the completion of this research. We extend our thanks to Prisca Blandin for the analysis of the ground-truth DNA mixtures, to the technicians and scientists in our lab for the DNA profile deconvolutions and to the volunteers for their participation.

References

[1] European Commission. Directorate General for Migration and Home Affairs., Deloitte Consulting., Advisory CVBA., Study on the feasibility of improving information exchange under the Prüm decisions: final report., Publications Office, LU, 2020. ⟨https://data.europa.eu/doi/10.2837/634009⟩ (Accessed 7 October 2024).
[2] D. Taylor, Using continuous DNA interpretation methods to revisit likelihood ratio behaviour, Forensic Sci. Int.: Genet. 11 (2014) 144–153, https://doi.org/10.1016/j.fsigen.2014.03.008.
[3] M.D. Coble, J.-A. Bright, Probabilistic genotyping software: an overview, Forensic Sci. Int.: Genet. 38 (2019) 219–224, https://doi.org/10.1016/j.fsigen.2018.11.009.
[4] P. Gill, C. Benschop, J. Buckleton, Ø. Bleka, D. Taylor, A review of probabilistic genotyping systems: EuroForMix, DNAStatistX and STRmix™, Genes 12 (2021) 1559, https://doi.org/10.3390/genes12101559.
[5] T. Hicks, F. Taroni, J. Curran, J. Buckleton, V. Castella, O. Ribaux, Use of DNA profiles for investigation using a simulated national DNA database: part II. Statistical and ethical considerations on familial searching, Forensic Sci. Int.: Genet. 4 (2010) 316–322, https://doi.org/10.1016/j.fsigen.2009.11.006.
[6] Ø. Bleka, M. Eduardoff, C. Santos, C. Phillips, W. Parson, P. Gill, Using EuroForMix to analyse complex SNP mixtures, up to six contributors, Forensic Sci. Int.: Genet. Suppl. Ser. 6 (2017) e277–e279, https://doi.org/10.1016/j.fsigss.2017.09.084.
[7] H. Kelly, Z. Kerr, K. Cheng, M. Kruijver, J.-A. Bright, Developmental validation of a software implementation of a flexible framework for the assignment of likelihood ratios for forensic investigations, Forensic Sci. Int.: Rep. 4 (2021) 100231, https://doi.org/10.1016/j.fsir.2021.100231.
[8] J. Hoogenboom, T. Sijen, C. Benschop, ProbRank: an efficient DNA database search method for complex mixtures per a quantitative likelihood ratio model, Forensic Sci. Int.: Genet. 65 (2023) 102884, https://doi.org/10.1016/j.fsigen.2023.102884.
[9] C.C.G. Benschop, L. Van De Merwe, J. De Jong, V. Vanvooren, M. Kempenaers, C. P. (Kees) Van Der Beek, F. Barni, E.L. Reyes, L. Moulin, L. Pene, H. Haned, T. Sijen, Validation of SmartRank: a likelihood ratio software for searching national DNA databases with complex DNA profiles, Forensic Sci. Int.: Genet. 29 (2017) 145–153, https://doi.org/10.1016/j.fsigen.2017.04.008.
[10] Ø. Bleka, M. Bouzga, A. Fonneløp, P. Gill, dnamatch2: an open source software to carry out large scale database searches of mixtures using qualitative and quantitative models, Forensic Sci. Int.: Genet. Suppl. Ser. 6 (2017) e404–e406, https://doi.org/10.1016/j.fsigss.2017.09.160.
[11] J.-A. Bright, D. Taylor, J. Curran, J. Buckleton, Searching mixed DNA profiles directly against profile databases, Forensic Sci. Int.: Genet. 9 (2014) 102–110, https://doi.org/10.1016/j.fsigen.2013.12.001.
[12] J.-A. Bright, D. Taylor, C. McGovern, S. Cooper, L. Russell, D. Abarno, J. Buckleton, Developmental validation of STRmix™, expert software for the interpretation of forensic DNA profiles, Forensic Sci. Int.: Genet. 23 (2016) 226–239, https://doi.org/10.1016/j.fsigen.2016.05.007.
[13] M. Zieger, S. Utz, A "forensic biobank" to establish comprehensive genetic frequency data for Switzerland, Forensic Sci. Int.: Genet. 40 (2019) 46–51, https://doi.org/10.1016/j.fsigen.2019.02.004.
[14] J. Buckleton, J. Curran, J. Goudet, D. Taylor, A. Thiery, B.S. Weir, Population-specific F values for forensic STR markers: a worldwide survey, Forensic Sci. Int.: Genet. 23 (2016) 91–100, https://doi.org/10.1016/j.fsigen.2016.03.004.
[15] Expert Working Group on Human Factors in Forensic DNA Interpretation, Forensic DNA Interpretation and Human Factors: Improving Practice Through a Systems Approach, National Institute of Standards and Technology, Gaithersburg, MD NIST IR 8503 (2024). ⟨https://doi.org/10.6028/NIST.IR.8503⟩ (Accessed 28 June 2024).
[16] J.-A. Bright, D. Taylor, Z. Kerr, J. Buckleton, M. Kruijver, The efficacy of DNA mixture to mixture matching, Forensic Sci. Int.: Genet. 41 (2019) 64–71, https://doi.org/10.1016/j.fsigen.2019.02.020.
[17] D. Taylor, D. Abarno, Using big data from probabilistic genotyping to solve crime, Forensic Sci. Int.: Genet. 57 (2022) 102631, https://doi.org/10.1016/j.fsigen.2021.102631.
[18] Loi fédérale sur l'utilisation de profils d'ADN dans les procédures pénales et sur l'identification de personnes inconnues ou disparues (Loi sur les profils d'ADN), 363, 20/06/2003.

[19] R. Puch-Solis, L. Rodgers, A. Mazumder, S. Pope, I. Evett, J. Curran, D. Balding, Evaluating forensic DNA profiles using peak heights, allowing for multiple donors, allelic dropout and stutters, Forensic Sci. Int.: Genet. 7 (2013) 555–563, https://doi.org/10.1016/j.fsigen.2013.05.009.

[20] T.R. Moretti, R.S. Just, S.C. Kehl, L.E. Willis, J.S. Buckleton, J.-A. Bright, D. A. Taylor, A.J. Onorato, Internal validation of STRmix™ for the interpretation of single source and mixed DNA profiles, Forensic Sci. Int.: Genet. 29 (2017) 126–144, https://doi.org/10.1016/j.fsigen.2017.04.004.

[21] A. Biedermann, S. Bozza, F. Taroni, Normative decision analysis in forensic science, Artif. Intell. Law 28 (2020) 7–25, https://doi.org/10.1007/s10506-018-9232-2.

[22] Federal Office of Police, CODIS in figures, (2024). ⟨https://www.fedpol.admin.ch/fedpol/en/home/sicherheit/personenidentifikation-neu/dna-und-codis/codis-zahlen.html⟩ (accessed September 25, 2024).

[23] P. Gill, T. Hicks, J.M. Butler, E. Connolly, L. Gusmão, B. Kokshoorn, N. Morling, R. A.H. Van Oorschot, W. Parson, M. Prinz, P.M. Schneider, T. Sijen, D. Taylor, DNA commission of the International society for forensic genetics: assessing the value of forensic biological evidence - Guidelines highlighting the importance of propositions, Forensic Sci. Int.: Genet. 36 (2018) 189–202, https://doi.org/10.1016/j.fsigen.2018.07.003.

[24] P. Gill, T. Hicks, J.M. Butler, E. Connolly, L. Gusmão, B. Kokshoorn, N. Morling, R. A.H. Van Oorschot, W. Parson, M. Prinz, P.M. Schneider, T. Sijen, D. Taylor, DNA commission of the International society for forensic genetics: assessing the value of forensic biological evidence - Guidelines highlighting the importance of propositions. Part II: Evaluation of biological traces considering activity level propositions, Forensic Sci. Int.: Genet. 44 (2020) 102186, https://doi.org/10.1016/j.fsigen.2019.102186.

[25] D. Taylor, J. Buckleton, Combining artificial neural network classification with fully continuous probabilistic genotyping to remove the need for an analytical threshold and electropherogram reading, Forensic Sci. Int.: Genet. 62 (2023) 102787, https://doi.org/10.1016/j.fsigen.2022.102787.